FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Antenatal Care & Pregnancy Monitoring

Tutorial | Week 1 · Foundations, ANC & early pregnancy | Obstetrics | Dr Nsingo

Antenatal care (ANC) is a schedule of planned contacts that screens for, prevents and treats conditions that kill mothers and babies. In Uganda that means early booking, at least 8 contacts (WHO 2016), routine HIV/syphilis/hepatitis B, Hb and blood-group testing, IPTp-SP, Td vaccination, iron–folic acid, deworming, an insecticide-treated net, BP and urine checks at every contact, danger-sign counselling and a birth plan.

Must know before the ward round

1. WHO 2016 and Uganda MoH (adopted 2018) recommend at least 8 contacts; first contact before 12 weeks, then 20, 26, 30, 34, 36, 38 and 40 weeks.
2. IPTp-SP = 3 tablets of sulfadoxine 500 mg/pyrimethamine 25 mg (1500/75 mg) by directly observed therapy from 13 weeks, at least 4 weeks apart, to delivery; aim for 3 or more doses. Never in the first trimester or with cotrimoxazole.
3. Iron–folic acid every day of pregnancy (30–60 mg elemental iron + 400 µg folic acid; ferrous sulphate 200 mg contains about 60 mg elemental iron). Anaemia is Hb below 11 g/dL, and severe anaemia is Hb below 7 g/dL.
4. Td (tetanus–diphtheria) 5-dose schedule. Td1 at first contact, Td2 at least 4 weeks later (and at least 2 weeks before delivery), Td3 at 6 months, then Td4 and Td5 a year apart.
5. Triple elimination means testing every woman for HIV, syphilis and hepatitis B at booking. HIV-negative women are retested in the third trimester, labour or postpartum. HIV-positive women start TLD the same day.
6. Measure BP and test urine protein at every contact, because pre-eclampsia is silent until it is severe. BP of 140/90 mmHg or more after 20 weeks needs urine protein and a plan the same day.
7. Deworm once after the first trimester (mebendazole 500 mg or albendazole 400 mg single dose). Consider low-dose aspirin 75 mg nightly from 12 weeks, and before 20 weeks, for women at high risk of pre-eclampsia.
8. Every contact ends with danger-sign counselling and an updated birth-preparedness plan (where, who, transport, money, blood donor, mama kit).
9. Booking tests are Hb, blood group and Rh, HIV, syphilis, HBsAg, urinalysis and an ultrasound before 24 weeks. Recheck Hb at 26–28 and 34–36 weeks.
10. GDM by 75 g OGTT at 24–28 weeks (WHO 2013): fasting 5.1 or more, 1 h 10.0 or more, or 2 h 8.5 or more mmol/L; one abnormal value is enough.
11. SFH in cm equals gestation in weeks ± 2 cm from 24 weeks. More than 3 cm off means ultrasound (dates, growth, liquor).
12. Triage every ANC attender for danger signs and BP before she joins the queue. BP 160/110 mmHg or more, or Hb below 7 g/dL, means act or refer the same day.

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1. Definitions and Key Concepts

Antenatal care is **not** a string of routine "check-ups". It is a planned programme of **screening, prevention, treatment and preparation** that targets the killers of Ugandan mothers and babies: **haemorrhage, hypertensive disorders, sepsis, obstructed labour, unsafe abortion, malaria, anaemia and HIV**, and for babies, **prematurity, birth asphyxia, stillbirth, congenital syphilis and vertical HIV/HBV**.

The WHO 2016 model, adopted by Uganda in 2018, calls these visits **contacts**, meaning a real interaction with a health worker, and recommends **at least 8**. The first should come **before 12 weeks**.

The one thing never to forget: **at every single contact, check the blood pressure and the urine, ask about danger signs, and update the birth plan.** Pre-eclampsia, anaemia and fetal growth restriction are silent until they are catastrophic, and ANC exists to find them early.

Term	Meaning
Antenatal care (ANC)	Care given by skilled health personnel to pregnant women and adolescent girls to secure the best health for mother and baby (WHO 2016)
Contact	Any active interaction between the woman and a health provider. The WHO chose this word over "visit" to stress quality, not just attendance
Booking	The first ANC contact, where the full history, examination, baseline tests and risk assessment are done
Focused / goal-oriented ANC (FANC)	The older WHO model (2002) of 4 targeted visits. Uganda's forms and much teaching still use this term
Gestational age (GA)	Weeks since the first day of the last normal menstrual period (LNMP). Term is 37+0 to 41+6 weeks, preterm is below 37 weeks and post-term is 42 weeks or more
EDD (Naegele's rule)	LNMP + 9 months + 7 days (= LNMP + 280 days). It assumes a 28-day cycle
High-risk pregnancy	A pregnancy with a factor that raises the risk of poor maternal or perinatal outcome, so the woman needs closer follow-up and delivery at a facility that can do caesarean section and transfusion (HC IV or hospital)
IPTp	Intermittent preventive treatment of malaria in pregnancy: full treatment doses of SP given at set intervals whether or not she has malaria
Triple elimination	Elimination of mother-to-child transmission of HIV, syphilis and hepatitis B through integrated ANC testing and treatment (Uganda MoH strategy from 2019)
Three delays	Delay in deciding to seek care, in reaching care, and in receiving care: the model that explains most Ugandan maternal deaths

The WHO 2016 eight-contact schedule

Contact	Gestation	Main extra tasks, beyond what is done at every contact
1	up to 12 weeks	Full booking history and exam; confirm and date the pregnancy (ultrasound before 24 weeks); Hb, blood group/Rh, HIV, syphilis, HBsAg, urinalysis; start iron–folic acid; Td1; LLIN; risk assessment
2	20 weeks	IPTp-SP (first dose from 13 weeks); deworming; anomaly/dating scan if not done; Td2
3	26 weeks	IPTp-SP; Hb recheck; GDM screen at 24–28 weeks if risk factors

Contact	Gestation	Main extra tasks, beyond what is done at every contact
4	30 weeks	IPTp-SP; fetal growth (SFH); counsel on labour signs
5	34 weeks	IPTp-SP; Hb; HIV retest if negative; plan the place of birth
6	36 weeks	Presentation. **Breech or transverse at 36 weeks means refer** for ECV or a planned CS; finalise the birth plan
7	38 weeks	Presentation and engagement; review the birth plan; IPTp-SP if due
8	40 weeks	Plan for **induction at 41 weeks** if still undelivered

2. Burden and Uganda Context

- **Maternal mortality ratio** is about **189 per 100,000 live births** (UDHS 2022). The main direct causes are PPH, hypertensive disorders, sepsis, abortion complications and obstructed labour. The main indirect causes are malaria, HIV and anaemia.
- **ANC coverage:** most women attend at least once and about **72% attend 4 or more times** (UDHS 2022). First-trimester booking remains low (about 30% in UDHS 2016), and completion of 8 or more contacts is very low (below 5% in several Ugandan studies) Verify *first-trimester booking figure is from UDHS 2016; update from UDHS 2022 if reported*.
- Why do women book late? Distance and transport costs, cost of "extras", not knowing they should come early, fear of HIV testing, needing the partner's permission, adolescent stigma, and cultural reluctance to reveal a pregnancy early.
- **Malaria in pregnancy** causes maternal anaemia, placental malaria, low birth weight, stillbirth and preterm birth. Primigravidae carry the highest burden because they have not yet developed antibodies against parasites that bind the placenta.
- **Anaemia** affects roughly a third or more of pregnant Ugandan women. Iron deficiency, malaria, hookworm, HIV and sickle cell disease all contribute.
- **HIV** affects roughly 1 in 20 ANC attenders, with wide regional variation (higher in fishing communities and some urban areas). ANC is the main entry point for eMTCT.

What each level of care can do

Level	ANC capacity
HC II	Health education, BP, urinalysis dipstick, IPTp, Td, iron–folate, LLIN, HIV/syphilis rapid tests. Refers risks
HC III	All of the above plus Hb (HemoCue), malaria RDT/microscopy, basic delivery services, ART
HC IV	All of the above plus blood grouping, theatre for **caesarean section**, blood transfusion, often ultrasound. Manages many high-risk women
General / regional referral hospital (e.g. Mengo, Mulago/Kawempe)	Obstetricians, ultrasound, OGTT, renal/liver tests, specialist clinics (hypertension, diabetes, cardiac), neonatal unit

UGANDA CONTEXT

Uganda adopted the 8-contact model in 2018, but many health centres still run the old 4-visit "focused ANC" pattern because of staffing and stockouts. On the ward you will see both. Whatever the number of visits, **every** contact should include BP, urine, fundal height and fetal heart, danger-sign counselling and IPTp if due. The Mother–Child Health Passport or ANC card is the woman's record: read it, write in it and teach her to bring it to every visit and to labour.

3. Principles and Framework

Understanding **why** each element exists makes the list easy to remember.

1. **Early dating.** Accurate gestational age decides everything later: whether a baby is preterm (steroids?), post-term (induction?), or small for dates. An ultrasound before 24 weeks, ideally a crown–rump length at 7–14 weeks, is accurate to within about a week. Fundal height and LNMP alone are much less reliable.
2. **Screening for silent disease.** Pre-eclampsia starts as abnormal placentation in the first half of pregnancy and shows as hypertension and proteinuria only after 20 weeks. **Measuring BP and urine protein at each contact** catches it before eclampsia. Anaemia and syphilis are also silent.
3. **Prevention by pharmacology:**
    - **Iron–folic acid.** Plasma volume rises about 45–50% but red cell mass only about 20–30%, so haemoglobin concentration falls. The fetus, placenta and expanded red cell mass need about 1 g of iron in total, far more than the diet of most Ugandan women provides. Folate prevents megaloblastic anaemia, and before about 28 days after conception it prevents neural tube defects.
    - **IPTp-SP.** The placenta sequesters *P. falciparum* (via VAR2CSA binding to chondroitin sulphate A), often without fever or peripheral parasitaemia. Giving a full treatment dose monthly clears silent placental infection and prevents new infection for several weeks. The result is less maternal anaemia and low birth weight.
    - **Td.** Neonatal tetanus comes from contaminating the cord stump. Maternal IgG antitoxin crosses the placenta and protects the newborn for the first weeks.
    - **Deworming.** Hookworm causes chronic blood loss and iron deficiency.
    - **Calcium** where dietary intake is low (1.5–2 g/day) and **aspirin** for high-risk women reduce pre-eclampsia. Aspirin shifts the prostacyclin–thromboxane balance and improves placentation, so it must start **before 20 weeks** (ideally from 12 weeks).
4. **Preventing vertical transmission:** ART to viral suppression (HIV), penicillin (syphilis, which crosses the placenta at any stage), tenofovir from 28 weeks for a high HBV viral load, plus the birth-dose vaccine.
5. **Birth preparedness.** The **"three delays"** model (delay in deciding to seek care, in reaching care, and in receiving care) explains most Ugandan maternal deaths. A birth plan and danger-sign knowledge attack delays 1 and 2.
6. **Risk is dynamic.** ANC is a cycle of **screen → classify risk → intervene → plan place of birth → re-screen** at every contact. The aim is to answer three questions every time: **Is she safe today? Is the baby growing and well? Is she still safe to deliver where she plans to?**

4. Application in Practice

For ANC the "patient" is usually a well woman, so the workup is a **structured screen**: a full booking workup at the first contact, then a shorter, targeted review at every contact after that.

Step 0: triage in the first 5 minutes

Before she sits in the ANC queue (often 40 women long at a Ugandan HC IV), someone must **triage** her.

1. **Ask about danger signs at the door**: bleeding, severe headache or blurred vision, fits, fever, leaking liquor, severe abdominal pain, difficulty breathing, reduced fetal movements, labour pains. Any "yes" means she skips the queue.
2. **Vital signs**: BP, pulse, temperature, respiratory rate; SpO_2 if a pulse oximeter is available.
3. **Act on the numbers at once:**
   - **BP 160/110 mmHg or more**, or 140/90 or more **with symptoms**: severe pre-eclampsia pathway ($MgSO_4$ loading dose, antihypertensive, refer). See the emergency flow in section 6.
   - **Pulse above 100/min at rest**, or **shock index (HR ÷ SBP) 0.9 or more**: look for bleeding, sepsis or severe anaemia.
   - **Temperature 38.0°C or more**: malaria RDT now; think pyelonephritis, chorioamnionitis.
   - **RR above 30/min or breathlessness at rest**: severe anaemia with heart failure, pneumonia, rheumatic heart disease.
4. **Call for help** (midwife in charge, medical officer) if any of the above. Otherwise proceed to the routine workup.

> **RED FLAG**
> Women die in ANC queues. A woman with a headache and BP 164/112 mmHg who waits three hours "for her turn" can fit on the bench. **Triage first, queue second.**

Step 1: focused history at booking (ask with purpose)

The booking history is the richest risk-screening tool you have. Each item is there because a certain answer changes the plan.

Question	Why it matters: the answer that changes the plan
Age	Under 18: pre-eclampsia, obstructed labour (small pelvis), anaemia, preterm birth. Over 35: chromosomal anomalies, hypertension, GDM, PPH. Either → high risk; hospital or HC IV delivery
LNMP, certainty, cycle length, contraception or breastfeeding at conception	Dating. An irregular cycle or conception on the pill or while breastfeeding makes LNMP unreliable → **dating scan before 24 weeks**. Gives the EDD and decides timing of IPTp, Td, OGTT
Gravidity, parity and each previous pregnancy in detail (year, place, GA, mode, weight, outcome, complications)	Previous **CS** (scar: deliver in hospital; how many; why; interval); **PPH** (recurs, so plan active management and IV access); **pre-eclampsia**; **stillbirth or neonatal death** (fresh or macerated; why?); **preterm birth**; **obstructed labour or fistula**; **macrosomia** (baby over 4 kg: GDM); **grand multiparity** (5 or more deliveries: PPH, rupture, malpresentation). Each moves her to hospital-level care
Abortions and ectopics	Cervical insufficiency (painless second-trimester losses), Rh sensitisation, fertility history
Current symptoms	Bleeding, pain, headache, visual disturbance, oedema, fever, dysuria, abnormal discharge, fetal movements (after 20 weeks) → treat today rather than "screen"
Medical history	Hypertension, diabetes, sickle cell disease, heart disease (rheumatic), asthma, epilepsy (valproate), TB, thyroid disease, previous transfusion → specialist clinic and drug review
HIV status and ART	Current regimen, adherence, last viral load; is the partner tested? Known positive on ART → **viral load at this contact**; on cotrimoxazole → **no SP**
Drugs and allergies	Sulfonamide allergy (SP and cotrimoxazole are contraindicated); teratogens (valproate, ACE inhibitors, warfarin, isotretinoin) → switch; efavirenz is now acceptable

Question	Why it matters: the answer that changes the plan
Previous Td doses (card)	Decides whether she needs Td1, the next dose, or none
Family history	Twins, diabetes, hypertension, sickle cell disease, congenital anomalies → targeted tests (OGTT, sickling)
Social history	Marital status and support, partner involvement, **intimate partner violence** (ask privately, alone), alcohol and smoking, occupation, adolescent in school, distance to facility and transport, money for delivery → birth plan and social support

At every later contact (2–3 minutes): How are you? **Danger signs? Fetal movements** (after 20 weeks)? Taking iron, ART and sleeping under the net? Any new symptoms (headache, swelling, dysuria, discharge, itching)? What doses are due today (read the card: IPTp, Td, Hb, HIV retest)?

Step 2: focused examination (Hutchison order, obstetric abdomen)

1. **Introduction and consent**: introduce yourself, confirm her identity, explain, gain consent and **offer a chaperone**.
2. **Hand hygiene.**
3. **Position and exposure**: ask her to empty her bladder; semi-recumbent (left tilt late in pregnancy to avoid supine hypotension); expose the abdomen from the xiphisternum to the symphysis, keeping her covered otherwise.
4. **General inspection**: well or unwell, distress, **pallor**, jaundice, hydration, nutrition. At booking: **height** (**below 150 cm** suggests a small pelvis and CPD risk), **weight and BMI**, gait (a limp or deformity suggests a contracted pelvis), spine.
5. **Vital signs**: **BP** (seated, correct cuff, arm at heart level; **140/90 mmHg or more** is hypertension, **160/110 mmHg or more** is severe), pulse, temperature, RR; **weight** at every contact.
6. **Hands**: palmar pallor; oedema of the hands.
7. **Face**: conjunctival and tongue pallor, jaundice, **facial oedema** (generalised oedema is more worrying than ankle oedema).
8. **Neck**: thyroid and lymph nodes (at booking).
9. **Chest and breasts** (at booking): **heart** (murmur of rheumatic heart disease), lungs; breasts as indicated.
10. **Obstetric abdominal examination (every contact)**:
    - **Inspection**: size, **scars** (Pfannenstiel = previous CS), linea nigra, striae, fetal movements.
    - **Fundal height**: before 24 weeks by landmarks (symphysis at 12 weeks, umbilicus at about 20–22 weeks); from 24 weeks **SFH in cm** ≈ gestation in weeks **± 2 cm** (up to 36 weeks). A difference of **more than 2–3 cm** means scan for growth, liquor or dates.
    - **Lie** (from about 32–36 weeks).
    - **Presentation** (from about 32–36 weeks).
    - **Engagement in fifths** near term.
    - **Auscultation of the fetal heart**: Doppler (from about 10–12 weeks) or Pinard (from about 20–24 weeks); normal **110–160 bpm**.
11. **Legs**: oedema (sacrum, ankles), varicose veins, calf tenderness.
12. **Pelvic / speculum**: **not routine**. Do it only if there is bleeding (and only after placenta praevia is excluded by scan), abnormal discharge, suspected ROM, or for cervical screening.
13. **Complete**: thank her, help her dress, **hand hygiene**, summarise and document.

Document: key positives (e.g. "pallor +, SFH 28 cm at 32 weeks") **and** key negatives (e.g. "no oedema, BP 118/72, no proteinuria").

Step 3: bedside tests (in the room, at most HC III and above)

Test	Normal	Act if
Urine dipstick (protein, glucose, nitrites, leucocytes)	Negative or trace	**Protein 1+ or more with BP 140/90 or more** → pre-eclampsia workup. **Glucose** on two occasions → RBS/OGTT. **Nitrites** → urine culture if possible and treat asymptomatic bacteriuria
HemoCue (point-of-care Hb)	11 g/dL or more	Below 11 → anaemia plan; **below 7 → admit/refer**
Dual HIV/syphilis rapid test	Both non-reactive	HIV reactive → confirm per national algorithm and **start TLD the same day**. Syphilis reactive → **treat today**
HBsAg rapid test	Negative	Positive → HBV plan (tenofovir assessment, birth dose within 24 h)
Malaria RDT	Negative	Any fever or anaemia → test; positive → treat (and it does not cancel IPTp later)
Urine hCG	–	Only if pregnancy is uncertain (early booking, unclear history)
Random blood sugar (glucometer)	–	Glycosuria, risk factors or symptoms: RBS **11.1 mmol/L or more** suggests diabetes; any doubt → formal FBS/OGTT (a glucometer screens, it does not diagnose)
Doppler FHR	110–160 bpm	Not heard when expected → ultrasound to confirm viability

Step 4: laboratory and imaging

Test	When	Expected / abnormal finding	How it changes management (and where it is available)
Hb / full blood count	Booking, **26–28 weeks**, **34–36 weeks**, and whenever pale	Hb 11 g/dL or more. Below 11 anaemia, **below 7 severe**. MCV low = iron deficiency; high = folate/B12	Therapeutic iron; look for cause; transfuse if severe. HemoCue at HC III; FBC at HC IV and hospital
Blood group and Rhesus	Booking	Rh-negative in a few percent of Ugandan women	Anti-D planning; donor for the birth plan. HC IV and hospital
Indirect Coombs (antibody screen)	Rh-negative women	Positive = already sensitised	Refer for fetal surveillance; anti-D no longer useful. Regional/national referral only
HIV (rapid, then confirmatory algorithm)	Booking; **retest if negative in the 3rd trimester**, labour or postpartum	Reactive	Same-day ART; eMTCT plan. All levels
Viral load	Women already on ART at first ANC; then per national schedule	Suppressed vs non-suppressed (cut-off varies by guideline: check the current Uganda protocol)	Non-suppressed → adherence support, high-risk infant prophylaxis. Sample taken at HC III/IV, processed centrally (hub system)
Syphilis (rapid treponemal or dual test; RPR/VDRL where available)	Booking; repeat in the 3rd trimester if high risk	Reactive	Benzathine penicillin; treat partner. RPR titre (hospital labs) helps judge activity and response
HBsAg	Booking	Positive	Tenofovir assessment; birth-dose vaccine within 24 h. HBV DNA/HBeAg only at referral centres

Test	When	Expected / abnormal finding	How it changes management (and where it is available)
Urinalysis ± urine culture	Dipstick every contact; culture if nitrites/leucocytes or symptoms	Bacteriuria	Treat (nitrofurantoin or amoxicillin per sensitivity) to prevent pyelonephritis and preterm labour. Culture mostly at regional/national referral
Malaria microscopy (blood slide)	Fever, anaemia, positive RDT needing parasite count	Parasites	Treat; severe malaria → IV artesunate. HC III and above
Stool microscopy	Unexplained or refractory anaemia	Hookworm ova	Deworm (after the 1st trimester). HC IV and hospital
Sickling test / Hb electrophoresis	Family history, recurrent crises, unexplained anaemia or jaundice	HbSS / HbSC	Sickle cell ANC clinic, folic acid 5 mg, malaria prevention plan. Regional/national referral
75 g OGTT (or FBS if OGTT not available)	**24–28 weeks** if risk factors or persistent glycosuria	GDM (WHO 2013): **fasting 5.1–6.9, 1 h 10.0 or more, 2 h 8.5–11.0 mmol/L** (any one value)	Diet, then metformin/insulin; growth scans; plan delivery. Mainly regional/national referral (Mengo, Mulago)
Obstetric ultrasound	**At least one before 24 weeks**; later if SFH discrepancy, malpresentation at 36 weeks, bleeding or no FHR	Dating, number, viability, placental site, anomalies, growth, liquor	Corrects the EDD, finds twins/praevia/FGR. Many HC IVs, all hospitals
Creatinine, LFTs, platelets	Raised BP or proteinuria; HIV baseline (creatinine for tenofovir)	Platelets below 100, raised ALT/AST, creatinine above about 90 µmol/L	Pre-eclampsia with severe features → admit and plan delivery. Hospital
Cervical screening (VIA or HPV test)	Usually offered postnatally; in pregnancy only if indicated or never screened	Acetowhite lesion / HPV positive	Referral pathway (treatment is usually deferred until after pregnancy). Available wherever the national screening programme runs; check your facility

In short: at booking do **Hb, group and Rh, HIV, syphilis, HBsAg, urinalysis and a scan before 24 weeks**; repeat **Hb at 26–28 and 34–36 weeks**; **retest HIV** in the 3rd trimester if negative; **urine and BP at every contact**; OGTT at 24–28 weeks if risk factors.

Step 5: the package at every contact (the "ANC checklist")

1. **Greet, confirm identity, and review the card or passport**: previous BP, Hb, tests done, doses given.
2. **Ask** how she is, about **danger signs** and fetal movements, and about adherence to iron, ART and bed-net use.
3. **Examine**: BP, weight, pallor, oedema, fundal height/SFH, lie and presentation, FHR.
4. **Test**: urine dipstick; Hb when due; retest HIV when due.
5. **Treat and prevent**: give due doses of IPTp-SP (directly observed), Td, iron–folic acid refills and deworming.
6. **Counsel**: danger signs, nutrition, birth preparedness, family planning after birth (postpartum IUD, implant), exclusive breastfeeding, and male involvement.
7. **Classify risk** and **decide the place of birth**. Refer if she is high-risk.
8. **Document** and **book the next contact**.

Step 6: preventive drugs and vaccines

Drug / vaccine	Dose	Route	Frequency / timing	Notes
Iron + folic acid (ferrous sulphate 200 mg ≈ 60 mg elemental iron + folic acid 400 µg)	1 tablet	Oral	**Daily** throughout pregnancy (continue about 3 months postpartum)	Take with vitamin C-rich food and **not with tea or milk**. Warn about black stools, nausea and constipation. Space it apart from calcium
Sulfadoxine–pyrimethamine (IPTp-SP)	**3 tablets** (sulfadoxine 500 mg / pyrimethamine 25 mg each = 1500/75 mg)	Oral, **DOT**	From **13 weeks** (2nd trimester), at each scheduled contact **at least 4 weeks apart**, until delivery. **Minimum 3 doses**	**Not** in the 1st trimester. **Not** in HIV-positive women on **cotrimoxazole**. Not with sulfa allergy. Can be taken on an empty stomach
Td (tetanus–diphtheria) vaccine	0.5 mL	IM (deltoid)	Td1 at first contact; Td2 **4 weeks** later (at least 2 weeks before EDD); Td3 6 months after Td2; Td4 1 year after Td3; Td5 1 year after Td4	Count previous documented doses. 5 doses protect through the childbearing years (probably longer)
Mebendazole	**500 mg**	Oral	**Single dose** after the 1st trimester	Albendazole 400 mg single dose is an alternative. Avoid in the 1st trimester
LLIN (long-lasting insecticidal net)	1 net	–	At the first contact	Sleep under it every night, and so should the baby
Calcium (where dietary calcium is low)	1.5–2.0 g elemental calcium per day	Oral	In **3 divided doses** at meals, from 20 weeks	WHO recommendation to prevent pre-eclampsia. Separate from iron by several hours. Availability is limited in Uganda
Low-dose aspirin (high risk of pre-eclampsia)	**75 mg** (many use 75–150 mg)	Oral	**Once daily at night**, start at 12 weeks (**before 20 weeks**) and continue to about 36 weeks or delivery	For previous pre-eclampsia, chronic hypertension, diabetes, renal or autoimmune disease, or 2 or more moderate risks (first pregnancy, age 40 or more, BMI 35 or more, twins, family history)

DRUG DOSING

IPTp-SP in one line: 3 tablets of SP (1500 mg/75 mg) swallowed in front of the health worker, **from 13 weeks, monthly (at least 4 weeks apart) until delivery, minimum 3 doses.** Contraindicated in the first trimester, in sulfa allergy, and in HIV-positive women taking cotrimoxazole. Routine iron–folic acid (400 µg) may continue. Avoid high-dose (5 mg) folic acid in the 2 weeks after SP because it can reduce SP efficacy.

Step 7: birth preparedness and complication readiness

Help her answer these, and write the answers down:

1. **Where** will she deliver? (Match the facility to her risk level.)
2. **Who** will accompany her? (A birth companion, and a decision-maker if the husband is away.)
3. **How** will she get there, day and night? (Boda boda number, ambulance, "saving for transport".)
4. **Money**: savings for transport and emergencies.
5. **Blood donor**: identify a potential donor, especially if she is high-risk.
6. **Mama kit / baby items**: sheets, gloves, cord ties, clean clothes, pads.
7. **Recognise labour and danger signs**, and know that she should come early if high-risk (e.g. previous CS: come at the onset of pains).

Step 8: follow-up and the postnatal link

- The ANC plan continues after birth: **postnatal contacts** (Uganda teaches "6 hours, 6 days, 6 weeks, 6 months"), with **family planning**, completion of Td doses, infant immunisation, EID for HIV-exposed infants, and cervical screening.

5. Clinical Acumen

Red flags

RED FLAG

Danger signs to teach every woman, who must come to hospital immediately:

- **Vaginal bleeding** at any time
- **Severe headache, blurred vision, epigastric pain or fits** (pre-eclampsia/eclampsia)
- **Fever** (malaria, pyelonephritis, chorioamnionitis)
- **Leaking fluid** from the vagina before labour (PROM)
- **Reduced or absent fetal movements**
- **Severe abdominal pain**
- **Swelling of the face and hands**
- **Difficulty breathing or extreme tiredness** (severe anaemia, heart failure)
- **Labour pains before 37 weeks**

Red flags for the clinician at a routine contact:

- BP 140/90 mmHg or more, or proteinuria 2+ or more: pre-eclampsia until proven otherwise.
- Hb below 7 g/dL, or anaemia with breathlessness or tachycardia: admit.
- SFH more than 3 cm below dates: suspect fetal growth restriction (FGR). More than 3 cm above dates: suspect twins, polyhydramnios, macrosomia or wrong dates.
- Malpresentation at 36 weeks or later, or an unengaged head at term in a primigravida (think CPD or placenta praevia).
- No fetal heart heard with a Doppler, or on ultrasound, when it should be.
- A history of previous CS **plus** a short interval, or 2 or more previous CS: hospital delivery.

Thinking like a clinician

- **Risk is dynamic.** A "low-risk" booking can become high-risk at 34 weeks. Re-assess at every contact and ask yourself: "Is this woman still safe to deliver at this level?"
- **If the SFH is small, think "Dates, Growth, Liquor":** wrong dates, FGR (pre-eclampsia, HIV, malaria, smoking, malnutrition), oligohydramnios (ruptured membranes?), or a transverse lie.
- **If the SFH is large, think "Dates, Twins, Liquor, Big baby, Fibroids":** wrong dates, multiple pregnancy, polyhydramnios (GDM, anomalies), macrosomia (GDM), fibroids.
- **If she has fever, think malaria first in Uganda**, but also UTI/pyelonephritis, chorioamnionitis, and HIV-related infections (TB, pneumonia). Always do an RDT or blood slide, even if she has had IPTp.
- **If she is anaemic, ask why.** Take a dietary history, check for malaria, deworm, ask about bleeding (haemorrhoids, APH), consider HIV and sickle cell, and look at the blood film. Treating with iron alone without finding the cause is a common student error.
- **A normal BP today does not mean normal tomorrow.** Teach the symptoms and schedule a recheck in 1 week if the BP was borderline.

- **Missed opportunities are a pattern.** A woman who books at 30 weeks has already missed IPTp doses, Td and early dating. Catch up **everything** at that one contact: test, give SP if 4 or more weeks since the last dose, give Td, and give iron and a net.
- **Pitfall:** giving SP to an HIV-positive woman on cotrimoxazole. She is already protected by cotrimoxazole, and combining two sulfa drugs increases the risk of severe skin reactions (Stevens–Johnson syndrome) and marrow toxicity.
- **Pitfall:** high-dose folic acid (5 mg) given with SP can antagonise pyrimethamine (both act on the folate pathway). The routine 400 µg dose is fine.

Common abnormal findings at ANC: the differentials

Finding	Conditions to consider	Distinguishing features	Key investigation
SFH small for dates	Wrong dates; FGR; oligohydramnios; transverse lie; IUFD	FGR: a risk factor (pre-eclampsia, HIV, malaria, smoking), slow serial growth. Oligohydramnios: fetal parts easily felt, ? PROM	Ultrasound: biometry, liquor, Doppler; review the LNMP
SFH large for dates	Wrong dates; twins; polyhydramnios; macrosomia; fibroids; molar pregnancy (early)	Twins: many fetal parts, two heads. Polyhydramnios: tense, fluid thrill, parts hard to feel. Fibroids: irregular firm masses	Ultrasound; RBS/OGTT for GDM
Raised BP	Gestational hypertension; pre-eclampsia; chronic hypertension; white-coat effect	Before 20 weeks means chronic. Proteinuria or organ dysfunction means pre-eclampsia	Urine protein, FBC (platelets), LFTs, creatinine; repeat BP after rest
Pallor / low Hb	Iron deficiency; malaria; hookworm; folate deficiency; HIV; sickle cell; blood loss	Iron: microcytic. Folate: macrocytic. Malaria: fever, splenomegaly. Sickle cell: history of crises, jaundice	FBC with indices, blood film, malaria RDT/slide, stool microscopy, sickling test / Hb electrophoresis
Glycosuria	Physiological (lower renal threshold in pregnancy); GDM; pre-existing diabetes	Physiological: once, low grade, normal RBS. GDM: persistent, risk factors, big baby, polyhydramnios	FBS/RBS; 75 g OGTT at 24–28 weeks
Malpresentation at 36 weeks	Breech; transverse/oblique lie	Breech: hard round head at the fundus. Transverse: head in a flank, empty pelvis	Ultrasound (also looks for placenta praevia, anomalies, fibroids)

Putting it together

Model one-line summary (booking):

> "Ms K.R., 19-year-old **primigravida** at **18 weeks** by LNMP (certain), booking today at HC IV. Asymptomatic, BP 112/70 mmHg, **pale**, SFH not yet applicable (fundus just below the umbilicus), FHR 146 bpm on Doppler. Booking bloods show **Hb 8.9 g/dL**, **syphilis reactive**, HIV non-reactive, HBsAg negative, O Rh-positive; urine **glucose 1+**, no protein."

Problem list:

1. Adolescent primigravida (pre-eclampsia, obstructed labour and anaemia risk).
2. Moderate anaemia (Hb 8.9 g/dL), cause not yet known.
3. Reactive syphilis test in pregnancy.
4. Single episode of glycosuria.
5. Preventive care due: IPTp-SP dose 1, Td, deworming, LLIN.

Working diagnosis: "Singleton intrauterine pregnancy at 18 weeks, **high-risk** (adolescent, moderate anaemia, syphilis): deliver at HC IV or hospital."

Worked example 1: interpreting the booking results

Result	Interpretation	Action today
Hb 8.9 g/dL	**Moderate** anaemia (7.0–9.9 g/dL)	Malaria RDT; mebendazole 500 mg (she is past the 1st trimester); therapeutic iron (**120 mg elemental + folic acid 400 µg daily**, i.e. ferrous sulphate 200 mg twice daily); FBC/film if available; recheck Hb in 4 weeks. Expect about 1 g/dL rise every 2–3 weeks
Syphilis reactive (rapid treponemal test)	Treponemal tests stay positive for life, so this could be old treated disease, but **in pregnancy, treat unless there is documented adequate treatment**	**Benzathine penicillin 2.4 MU IM weekly × 3** (unknown duration); treat the partner; RPR titre if the lab has it; tell the delivery team to examine and treat the baby
Glucose 1+ once	The renal threshold for glucose falls in pregnancy, so a single trace/1+ is often physiological	Repeat dipstick at the next contact; RBS; OGTT at 24–28 weeks if it persists or she has risk factors
HIV non-reactive	Negative now, but she is still at risk	**Retest in the 3rd trimester** (or labour/postpartum)

Then give **IPTp-SP dose 1 (DOT)**, Td (count previous doses), and an LLIN; book the next contact in 4 weeks (for the Hb recheck and IPTp dose 2).

Worked example 2: interpreting an OGTT at 26 weeks

A 34-year-old G4P3 with a previous 4.3 kg baby has a 75 g OGTT: **fasting 5.4 mmol/L, 1 h 9.6, 2 h 8.0.**

- Fasting 5.4 is **5.1 or more**, so this is **GDM**. **One abnormal value is enough**; the normal 1 h and 2 h values do not cancel it.
- If the fasting value had been **7.0 or more** (or the 2 h value **11.1 or more**), this would be **diabetes in pregnancy** (overt diabetes), not GDM, and she would need specialist care and probably insulin from the start.
- Plan: diabetes/ANC clinic referral, diet and exercise, glucose monitoring; metformin or insulin if targets are not met; growth scan; deliver in hospital. See the diabetes-in-pregnancy topic.

> CLINICAL PEARL
> Present an ANC patient in the order **dating → risk factors → today's findings → results → what is due today → place of birth**. A senior can then check in 30 seconds that nothing has been missed.

Model presentation (ANC review)

> "Mrs A.N. is a **24-year-old G2P1+0** at **32 weeks + 4 days** by a 10-week scan, attending her **fourth ANC contact**. She has **no danger signs** and reports good fetal movements. She is HIV-negative (retest due), VDRL non-reactive, HBsAg negative and **O Rh-positive**. Booking Hb was 10.4 g/dL and today it is 11.2 g/dL on iron–folate. She has had **2 doses of IPTp** (last 5 weeks ago) and **Td2**. On examination BP is 118/72 mmHg, there is no pallor or oedema, SFH is 32 cm, and she has a **singleton, longitudinal lie, cephalic, 5/5 palpable**, with FHR 140 bpm. Urine is clear. **Summary: a low-risk pregnancy progressing well.** Plan: IPTp dose 3 today, HIV retest, continue iron–folate, birth plan at Mengo, next contact at 34 weeks."

6. Common Problems and Their Management

Anaemia

Severity (WHO)	Hb	What to do
Mild	10.0–10.9 g/dL	Therapeutic iron, deworm, test for malaria, give dietary advice, recheck Hb in 4 weeks
Moderate	7.0–9.9 g/dL	As above. Look for the cause (film, malaria, stool, HIV, sickling). Treat. Plan delivery at HC IV or hospital
Severe	**below 7 g/dL**	**Admit/refer to a facility with blood.** Transfuse if Hb is below 7 g/dL near term (from about 36 weeks) or with signs of heart failure/decompensation, or if Hb is below 5 g/dL at any gestation. Transfuse slowly (packed cells). Consider furosemide 20–40 mg IV between units if she is at risk of fluid overload

- **Therapeutic iron:** WHO recommends **120 mg elemental iron + 400 µg folic acid daily** until Hb normalises, which is ferrous sulphate 200 mg **twice daily**. Some Ugandan protocols use ferrous sulphate 200 mg three times daily with folic acid 5 mg daily; if so, remember the SP–folate timing. Continue for about 3 months after Hb normalises to refill stores. Expect Hb to rise by about **1 g/dL every 2–3 weeks** if she is taking it.
- **Treat the cause:** malaria (artemether–lumefantrine in the 2nd and 3rd trimesters; quinine or artemether–lumefantrine in the 1st trimester per current guidance), deworming, and HIV care.
- **Parenteral iron** (e.g. iron sucrose or ferric carboxymaltose, where available at referral hospitals) is used if she is intolerant of or not responding to oral iron, or presents late in the third trimester with moderate anaemia.

Raised blood pressure

1. Repeat after 15 minutes of rest, with the correct cuff.
2. **140/90 mmHg or more after 20 weeks:** dipstick for protein, and ask about headache, visual symptoms and epigastric pain.
3. **Gestational hypertension without proteinuria and without symptoms:** refer to the hospital clinic. Start oral **methyldopa 250 mg 8-hourly** (up to 500 mg–1 g 8-hourly) or **nifedipine retard 20 mg 12-hourly** if BP is persistently 140/90 mmHg or more (NICE/CHAP threshold, as in the pre-eclampsia topic; some older protocols wait until 150/100). Review weekly with BP and urine.
4. **Pre-eclampsia** (hypertension + proteinuria or organ dysfunction): **admit to hospital**.
5. **Severe features** (BP 160/110 mmHg or more, or symptoms): **$MgSO_4$ loading dose** and antihypertensive, then **refer urgently** (see the pre-eclampsia topic).

HIV-positive at ANC (eMTCT)

1. Post-test counselling. **Start ART the same day** (preferred first line **TLD**: tenofovir 300 mg / lamivudine 300 mg / dolutegravir 50 mg, **one tablet once daily**). Start **cotrimoxazole** prophylaxis per national criteria, which means **no SP**.
2. Baseline tests (as available): CD4 or TB screen, creatinine, HBsAg, and viral load per schedule. For a woman already on ART, do a **viral load at the first ANC contact**.
3. Adherence counselling, partner testing (index testing), and disclosure support.
4. Plan the delivery: facility birth, **infant prophylaxis** (nevirapine ± zidovudine according to the infant's risk), early infant diagnosis (DNA PCR at 6 weeks), and infant feeding counselling (exclusive breastfeeding while on ART).

Syphilis-positive

Stage	Drug	Dose	Route	Frequency
Early (primary, secondary, early latent under 2 years)	**Benzathine penicillin G**	**2.4 million units** (split as 1.2 MU in each buttock)	Deep IM	**Single dose**
Late latent or unknown duration (most ANC cases)	Benzathine penicillin G	2.4 million units	Deep IM	**Weekly × 3 doses**
Penicillin allergy	Erythromycin 500 mg 6-hourly × 14 days (does not treat the fetus reliably, so the baby must be treated at birth) or ceftriaxone per protocol	–	Oral / IM	–

Treat the partner(s), and examine and treat the baby at birth.

Hepatitis B-positive (HBsAg+)

- Assess for **HBV DNA** if available. High risk (HBV DNA **200,000 IU/mL or more**, or HBeAg positive) means **tenofovir disoproxil fumarate 300 mg daily from 28 weeks** (Uganda practice: from 28–32 weeks) until at least delivery, then review.
- The baby gets **hepatitis B birth-dose vaccine within 24 hours** (and HBIG where available), then the routine pentavalent series.
- An HIV/HBV co-infected woman is already covered by TLD, which contains tenofovir and lamivudine.

Rhesus-negative

- Check the partner's group if possible, and check for antibodies (indirect Coombs test) if available.
- **Anti-D immunoglobulin** within **72 hours** of any sensitising event (bleeding, abortion, ectopic, ECV, trauma) and after delivery of an Rh-positive baby (e.g. **1500 IU (300 µg) IM**). Routine antenatal prophylaxis at 28 weeks is standard in the UK but rarely available in Uganda.

Malpresentation at 36 weeks

- Confirm with ultrasound. Refer to hospital for **external cephalic version** (ECV) if there are no contraindications, or plan a **caesarean section** at 39 weeks. **Never plan a home or HC III delivery for a breech or transverse lie.**

Post-dates

- At 40 weeks, confirm dates and give counselling. At **41 weeks**, offer **induction of labour** (WHO recommends induction at 41 weeks or later in a low-risk pregnancy). This needs hospital/HC IV admission.

High-risk criteria that require delivery at HC IV or hospital

Past obstetric	Current pregnancy	Medical / social
Previous CS or uterine surgery	Pre-eclampsia or hypertension	Diabetes, heart disease, sickle cell disease, HIV with a detectable VL
Previous PPH or retained placenta	Multiple pregnancy	Age below 18 or above 35 years
Previous stillbirth or neonatal death	Malpresentation after 36 weeks	Height below 150 cm
Grand multiparity (5 or more deliveries)	APH	Severe anaemia (Hb below 7 g/dL)

Past obstetric	Current pregnancy	Medical / social
Previous obstructed labour, fistula or 3rd/4th degree tear	Post-dates (41 weeks or more), PROM	Previous infertility or ART
3 or more consecutive miscarriages	Polyhydramnios, FGR	Primigravida (many Ugandan units advise facility delivery for all)

Emergency flow: a woman arrives at the ANC clinic with a danger sign

1. **Call for help** and move her to the examination couch. Check **airway, breathing and circulation**: BP, pulse, RR, SpO_2, temperature, consciousness.
2. **Is she fitting, or is BP 160/110 mmHg or more with symptoms?** Put her in the left lateral position, give oxygen, give **$MgSO_4$ 4 g IV over 5–10 minutes + 10 g IM (5 g in each buttock)** per the Pritchard regimen, and treat BP (nifedipine or hydralazine). Refer urgently.
3. **Is she bleeding? Do not do a digital vaginal exam** (placenta praevia). Insert 2 large-bore IV cannulae, send blood for grouping and cross-match, give crystalloid, and refer to a CS-capable facility.
4. **Is she febrile?** Do a malaria RDT, check for UTI/chorioamnionitis, and treat (e.g. artesunate IV for severe malaria).
5. **Reduced fetal movements?** Auscultate the FHR with a Doppler or Pinard. If absent, confirm with ultrasound and manage as IUFD. If present but abnormal, refer for monitoring.
6. **Write a referral note**, send a health worker with her if possible, and phone ahead.

What happens when ANC is missed or poor

Missed element	Consequence
No BP/urine screening	Eclampsia, abruption, HELLP, stroke, stillbirth
No iron / missed anaemia	Severe anaemia leads to death from moderate PPH, heart failure and poor wound healing
No IPTp or LLIN	Placental malaria, LBW, stillbirth, maternal anaemia, cerebral malaria
No HIV test / no ART	Vertical HIV transmission (about 15–45% without intervention; below 1–2% with suppressed VL and breastfeeding on ART)
No syphilis test	Stillbirth, congenital syphilis, neonatal death
No Td	Neonatal tetanus
No dating scan	Unnecessary or missed induction; wrong steroid decisions
No birth plan	Home delivery, the three delays, PPH or obstructed labour far from help

Drug side effects to know: iron causes GI upset, constipation and black stools. SP can cause rare severe cutaneous reactions (SJS/TEN), so stop and never re-challenge. Td causes local pain and swelling.

7. Guidelines, Policy and Law in Uganda

- **Uganda Clinical Guidelines 2023, Section 16.1 (Antenatal Care)** and the **Essential Maternal and Newborn Clinical Care Guidelines** set the national ANC package.
- **WHO 2016 ANC model** (at least 8 contacts) was **adopted by Uganda in 2018**; many facilities still run the 4-visit FANC pattern.
- **Triple elimination** (HIV, syphilis, hepatitis B) has been the MoH strategy since **2019**: test every woman at booking with the dual HIV/syphilis test and HBsAg.

- **Consolidated HIV guidelines (2022)**: same-day TLD for HIV-positive women; viral load at the first ANC contact for women already on ART; cotrimoxazole per national criteria.
- **Mother–Child Health Passport / ANC card**: the woman's own record; write in it at every contact.
- **Td has replaced TT** in the national schedule. Many clinics give a free **LLIN** at the first visit.

UGANDA CONTEXT

Uganda's malaria-in-pregnancy guidance explicitly states that **HIV-positive mothers on cotrimoxazole should not be given SP**. Cotrimoxazole (960 mg daily) already gives malaria protection. Uganda has switched from TT to **Td** vaccine. Many Ugandan ANC clinics give women a free **LLIN** at the first visit.

UGANDA CONTEXT

Where Uganda and international guidance diverge:

- **Hepatitis B**: Uganda practice gives tenofovir from **28–32 weeks** to women with a high HBV viral load or HBeAg positivity. Where HBV DNA and HBeAg testing are not available, **WHO (2024)** now advises tenofovir prophylaxis for **all** HBsAg-positive pregnant women; follow the current Uganda hepatitis B protocol at your facility Verify *whether Uganda has adopted the WHO 2024 test-free tenofovir recommendation*.
- **Antihypertensive threshold**: NICE/CHAP treat persistent BP of **140/90 mmHg or more**; some older Ugandan protocols wait until **150/100**.
- **Therapeutic iron**: WHO uses 120 mg elemental iron + 400 µg folic acid daily; some Ugandan protocols use ferrous sulphate 200 mg three times daily with folic acid 5 mg (mind the SP–folate interaction).
- **Aspirin dose**: 75 mg is common in Uganda and the UK; ACOG and many others use **75–150 mg** (often 150 mg) Verify *dose stated in UCG 2023*.
- **Anti-D**: routine antenatal prophylaxis at 28 weeks (UK standard) is rarely available in Uganda; give it after sensitising events and delivery.

8. Key Points

High-yield points

- At **every** contact: **BP, urine, danger signs, birth plan** (plus SFH and FHR).
- **At least 8 contacts**, first **before 12 weeks**.
- **Triage first, queue second.**
- **Booking tests**: Hb, group and Rh, HIV, syphilis, HBsAg, urinalysis, scan before 24 weeks.
- **IPTp-SP** from 13 weeks, monthly, DOT, ≥3 doses; never in the first trimester or with cotrimoxazole.
- **Iron–folic acid daily**; anaemia is Hb <11, severe <7 (admit).
- **Td** 5-dose schedule; **mebendazole** once after the first trimester; **LLIN** at the first contact.
- **Aspirin** for high-risk women must start **before 20 weeks**.
- **Syphilis reactive in pregnancy: treat today.**
- **Malpresentation at 36 weeks → refer**; **41 weeks → offer induction**.

Numbers & doses

Item	Value
ANC contacts (WHO 2016, Uganda 2018)	**At least 8**: before 12 weeks, then 20, 26, 30, 34, 36, 38, 40 weeks
Dating scan	At least one ultrasound **before 24 weeks**

Item	Value
EDD (Naegele)	LNMP **+ 9 months + 7 days** (280 days)
IPTp-SP	**3 tablets** (1500 mg/75 mg), **DOT**, from **13 weeks**, **at least 4 weeks apart**, **3 or more doses**
Routine iron–folic acid	30–60 mg elemental iron (ferrous sulphate 200 mg ≈ 60 mg) + **folic acid 400 µg daily**
Therapeutic iron (anaemia)	**120 mg elemental iron daily** + folic acid 400 µg; expect Hb rise of about 1 g/dL every 2–3 weeks
Anaemia (WHO)	Hb **below 11 g/dL**; mild 10.0–10.9; moderate 7.0–9.9; **severe below 7**
Transfusion	Hb <7 near term (≥36 weeks) or decompensated; Hb <5 at any gestation; furosemide 20–40 mg IV between units if at risk of overload
Hb checks	Booking, **26–28 weeks**, **34–36 weeks**
Td schedule	Td1 first contact; Td2 **4 weeks** later (and 2 weeks or more before delivery); Td3 6 months; Td4 and Td5 1 year apart
Mebendazole	**500 mg single dose** after the 1st trimester (or albendazole 400 mg)
Aspirin (high risk of pre-eclampsia)	**75 mg nightly** (75–150 mg) from **12 weeks**, start **before 20 weeks**, continue to about 36 weeks
Calcium (low-intake populations)	**1.5–2 g elemental daily** in divided doses from 20 weeks
Hypertension / severe	**140/90** / **160/110 mmHg**
Methyldopa / nifedipine retard	**250 mg 8-hourly** (up to 500 mg–1 g 8-hourly) / **20 mg 12-hourly**
$MgSO_4$ (Pritchard)	**4 g IV over 5–10 min + 10 g IM** (5 g each buttock)
Shock index	HR ÷ SBP **0.9 or more** = look for bleeding, sepsis, anaemia
SFH	cm ≈ weeks **± 2 cm** from 24 weeks; discrepancy more than 3 cm → scan
FHR	**110–160 bpm**
GDM (WHO 2013, 75 g OGTT at 24–28 weeks)	Fasting **5.1**, 1 h **10.0**, 2 h **8.5 mmol/L** (any one value); overt diabetes fasting ≥7.0 or 2 h ≥11.1
Syphilis	Benzathine penicillin **2.4 MU IM**; single dose early, **weekly × 3** if late/unknown
HBV	Tenofovir **300 mg daily from 28 weeks** if HBV DNA ≥200,000 IU/mL or HBeAg+; birth dose within **24 h**
Anti-D after sensitising event	Within **72 hours** (Uganda vial usually 1500 IU / 300 µg IM)
Malpresentation	Refer at **36 weeks** for ECV or planned CS (CS at 39 weeks)
Post-dates	Offer induction at **41 weeks**
Height suggesting CPD risk	**Below 150 cm**
Postnatal contacts	**6 hours, 6 days, 6 weeks, 6 months**

Exam traps

- **"Give SP to every pregnant woman from 13 weeks."** → Not if she is **HIV-positive on cotrimoxazole**, allergic to sulfa, or in the **first trimester**. Cotrimoxazole already protects her, and two sulfa drugs together raise the risk of SJS.
- **"She had IPTp, so the fever cannot be malaria."** → IPTp reduces but does not abolish malaria. **Every febrile pregnant woman needs an RDT or slide.**

- **"A reactive rapid syphilis test may be an old infection, so wait for the RPR."** → In pregnancy, **treat the same day** unless adequate previous treatment is documented. Waiting loses the woman and the baby.
- **"Glycosuria at ANC = gestational diabetes."** → The renal glucose threshold falls in pregnancy; one episode is often physiological. **Diagnose GDM only with an OGTT/FBS.**
- **"Start aspirin when BP rises at 28 weeks."** → Aspirin works by improving placentation and must start **before 20 weeks** (ideally 12). Starting at 28 weeks does little.
- **"Treat anaemia with iron and send her home."** → Always **look for the cause**: malaria, hookworm, HIV, sickle cell, bleeding, diet. Hb below 7 g/dL means **admit/refer**.
- **"She booked at 30 weeks, so skip the booking tests."** → Late booking is a reason to do **everything at once**: all booking tests, SP if due, Td, iron, LLIN, and dating by scan (less accurate now, but still useful).
- **"Do a VE at booking to assess the cervix."** → Pelvic examination is **not routine** at ANC; and **never** a digital VE in a woman who is bleeding until placenta praevia is excluded.

Questions seniors ask

Viva questions

1. **Why do we give IPTp even to women without malaria?** Because placental malaria is often asymptomatic, with parasites sequestered in the placenta and not seen in the peripheral blood. Monthly full treatment clears it and gives post-treatment prophylaxis. This reduces anaemia, LBW and perinatal death.
2. **Why not SP in the first trimester?** Pyrimethamine is a folate antagonist (it inhibits dihydrofolate reductase), so there is a theoretical risk to organogenesis. It is also unnecessary before placental sequestration becomes significant.
3. **Why 8 contacts rather than 4?** Evidence that 4 visits were associated with more perinatal deaths than 8 or more, plus more third-trimester contacts to catch pre-eclampsia, FGR and malpresentation. Women's experience of care is also better.
4. **What is the purpose of the 36-week contact?** To confirm presentation (breech or transverse means referral for ECV or CS) and to finalise the place of birth.
5. **How do you manage a woman with Hb 6.5 g/dL at 37 weeks?** Admit, look for the cause, transfuse packed cells slowly (with furosemide if she is at risk of overload), treat malaria/worms, give therapeutic iron, deliver in hospital with active management of the third stage, and continue iron postpartum.

On the ward round

1. **What are the booking investigations, and why each?** Hb (anaemia), group and Rh (anti-D, blood planning), HIV (same-day ART), syphilis (prevent stillbirth and congenital syphilis), HBsAg (birth dose and tenofovir), urinalysis (protein, glucose, bacteriuria) and a scan before 24 weeks (dating, number, placenta).
2. **Which three things do you check at every contact, whatever else happens? BP, urine protein and danger signs**, plus fetal growth (SFH) and FHR. Pre-eclampsia is silent until it is severe.
3. **This woman's SFH is 27 cm at 32 weeks. What next?** It is more than 3 cm below dates: think **"Dates, Growth, Liquor"**. Check the dating, look for risk factors (BP, HIV, malaria, smoking), and arrange an **ultrasound** for biometry, liquor and (if available) Doppler.
4. **When would you not give IPTp-SP today?** Gestation below 13 weeks, a dose less than 4 weeks ago, sulfa allergy, or HIV-positive on cotrimoxazole.
5. **An HIV-positive woman on TLD books at 16 weeks. What extra do you do? Viral load now**, adherence and disclosure check, cotrimoxazole per national criteria (then no SP), HBsAg, creatinine if available, partner testing, and a facility-delivery and infant prophylaxis plan.
6. **Who needs delivery at HC IV or hospital?** Previous CS or PPH, grand multipara, pre-eclampsia, twins, malpresentation at 36 weeks, age below 18 or above 35, height below 150 cm, severe anaemia, and medical disease (diabetes, heart disease, sickle cell, HIV with detectable VL).

7. **What is in a birth plan?** Where, who comes with her, how she will travel (day and night), money, a blood donor, the mama kit, and knowing labour and danger signs.

Mnemonics

Booking tests: "Triple H + GUSS"

- **H**IV
- **H**b (full blood count)
- **H**BsAg (hepatitis B)
- **G**roup and Rhesus
- **U**rinalysis (protein, glucose, nitrites)
- **S**yphilis
- **S**can (ultrasound before 24 weeks)

Preventive package: "FIT MEN"

- **F**olic acid + iron (FeFo)
- **I**PTp-SP
- **T**d vaccine
- **M**ebendazole
- **E**ducation (danger signs, birth plan, nutrition, FP)
- **N**et (LLIN)

Danger signs: "B-FLASH"

- **B**leeding
- **F**ever
- **L**eaking liquor
- **A**bdominal pain (severe)
- **S**welling or **S**evere headache / blurred vision / fits
- **H**ypo-activity of the baby (reduced movements)

The 8 contacts: "12, 20, 26, 30, 34, 36, 38, 40". Remember it as "one in the first, two in the second (20, 26), five in the third (30, 34, 36, 38, 40)".

Small SFH: "Dates, Growth, Liquor"; large SFH: "Dates, Twins, Liquor, Big baby, Fibroids".

OSCE tips

> EXAM FAVOURITE
> Examiners love the question "**What would you do at the first ANC visit?**" Structure your answer as: **History, Examination, Investigations, Interventions (drugs and vaccines), Counselling, Classification of risk, Next appointment.** Say "**at every contact I check BP and urine**" and you will score.

- In a counselling station, use the woman's language level. Check her understanding by asking her to repeat back the danger signs ("teach-back").
- When giving SP, say aloud "**directly observed**", confirm the gestation is 13 weeks or more, that it has been 4 weeks or more since the last dose, that she is not on cotrimoxazole, and that she has no sulfa allergy.
- Always wash your hands, introduce yourself, gain consent, and offer a chaperone.
- In the obstetric abdominal examination, say each step in order: inspection, fundal height (SFH), lie, presentation, engagement in fifths, fetal heart.

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	How many ANC contacts does WHO (2016) recommend, and at what gestations?	**At least 8**: first contact before **12 weeks**, then **20, 26, 30, 34, 36, 38 and 40 weeks** (1 in the first trimester, 2 in the second, 5 in the third). Uganda adopted this in 2018.
2	IPTp-SP: dose, start, interval and minimum number of doses	**3 tablets of SP** (1500 mg sulfadoxine / 75 mg pyrimethamine), **directly observed**, from **13 weeks**, at each contact **at least 4 weeks apart**, until delivery. **At least 3 doses.**
3	Three contraindications to IPTp-SP	1. **First trimester** (folate antagonist) 2. **HIV-positive woman on cotrimoxazole** (already protected; additive sulfa toxicity) 3. **Sulfonamide allergy** (risk of SJS/TEN)
4	Routine iron–folic acid in pregnancy (WHO)	**30–60 mg elemental iron + 400 µg folic acid daily** throughout pregnancy. Ferrous sulphate 200 mg ≈ 60 mg elemental iron. Prefer 60 mg where anaemia prevalence is 40% or more.
5	Definition and grading of anaemia in pregnancy (WHO)	Anaemia is **Hb below 11 g/dL**. Mild 10–10.9, moderate 7–9.9, **severe below 7 g/dL**.
6	Td vaccination schedule for a woman with no prior doses	Td1 at first contact, **Td2 at least 4 weeks later** (and at least 2 weeks before delivery), Td3 6 months after Td2, Td4 1 year later, Td5 1 year later. **0.5 mL IM.** 5 doses protect through the childbearing years (probably longer).
7	Deworming in pregnancy	**Mebendazole 500 mg** single dose (or albendazole 400 mg single dose) **after the first trimester**.
8	Booking investigations (list them)	**Hb/FBC, blood group and Rh, HIV, syphilis, HBsAg, urinalysis**, plus an **ultrasound before 24 weeks**. Malaria test if febrile; OGTT at 24–28 weeks if risk factors.
9	What do you do at EVERY ANC contact?	Ask about **danger signs** and fetal movements; measure **BP**; **urine dipstick** for protein; weight; pallor/oedema; **fundal height/SFH**; lie and presentation; FHR; give due drugs (IPTp, Td, FeFo); counsel; update the **birth plan**; classify risk.
10	SFH rule of thumb and when to act	From 24 weeks, **SFH in cm ≈ gestational age in weeks ± 2 cm**. A difference of more than 2–3 cm means an ultrasound for dates, growth and liquor.
11	Danger signs to teach every pregnant woman	Vaginal **bleeding**; **severe headache/blurred vision/fits**; **fever**; **leaking fluid**; **reduced fetal movements**; **severe abdominal pain**; swelling of the face and hands; breathlessness; labour pains before 37 weeks.
12	Who should get low-dose aspirin in pregnancy, and how?	Women at **high risk of pre-eclampsia** (previous pre-eclampsia, chronic HTN, diabetes, renal/autoimmune disease, or 2 or more moderate risks). **Aspirin 75 mg (75–150 mg) nightly**, started from **12 weeks and before 20 weeks**, until about 36 weeks or delivery.
13	WHO calcium supplementation in pregnancy	Where dietary calcium is low: **1.5–2.0 g elemental calcium daily in 3 divided doses** with meals, to reduce pre-eclampsia. Separate from iron by several hours.

#	Question	Answer
14	Syphilis in pregnancy: treatment	Early (under 2 years): **benzathine penicillin G 2.4 MU IM single dose**. Late latent/unknown duration: **2.4 MU IM weekly × 3**. Treat the partner; evaluate and treat the baby.
15	HBsAg-positive pregnant woman: key actions	Assess HBV DNA if possible. If **200,000 IU/mL or more** (or HBeAg+), give **tenofovir 300 mg daily from 28 weeks**. Baby: **HepB birth dose within 24 hours** (plus HBIG where available).
16	Newly diagnosed HIV at ANC: first steps	Post-test counselling; **start TLD the same day** (TDF 300/3TC 300/DTG 50 once daily); cotrimoxazole per criteria (so **no SP**); baseline tests; partner testing; adherence support; plan infant prophylaxis and EID.
17	The 'three delays' model	Delay 1: **deciding** to seek care. Delay 2: **reaching** care. Delay 3: **receiving** adequate care at the facility. The birth plan and danger-sign counselling target delays 1 and 2.
18	Components of a birth-preparedness plan	Place of birth; skilled attendant; birth companion; transport (day and night); money saved; blood donor; mama kit and baby items; knowing danger and labour signs; a decision-maker if the partner is away.
19	Why is SP given monthly rather than once?	Each dose clears existing (often silent, placental) parasites and gives about 4–6 weeks of **post-treatment prophylaxis**, so monthly dosing keeps the placenta protected through the second and third trimesters.
20	Malpresentation found at the 36-week contact: what now?	Confirm with **ultrasound** (also check placenta and anomalies). **Refer to hospital** for **ECV** if no contraindication, or plan **CS**. Never plan a lower-level or home delivery.
21	When is Hb checked in a routine pregnancy, and what Hb means admit/refer?	At **booking**, **26–28 weeks** and **34–36 weeks** (and whenever she is pale). Anaemia is Hb **below 11 g/dL**; **below 7 g/dL** is severe: **admit or refer to a facility with blood**.
22	75 g OGTT thresholds for GDM (WHO 2013), and when to do it	At **24–28 weeks** if risk factors or persistent glycosuria. GDM if **fasting 5.1–6.9**, **1 h 10.0 or more**, or **2 h 8.5–11.0 mmol/L**: **one abnormal value is enough**. Fasting 7.0 or more (or 2 h 11.1 or more) = diabetes in pregnancy (overt).
23	A booking rapid syphilis test is reactive. What do you do today?	**Treat today** unless adequate previous treatment is documented (treponemal tests stay positive for life, but waiting risks stillbirth and congenital syphilis). **Benzathine penicillin 2.4 MU IM weekly × 3** if duration unknown (single dose if early). Treat the partner; RPR titre if available; alert the delivery team to examine/treat the baby.
24	Triage at the ANC clinic: what must happen before a woman joins the queue?	Ask about **danger signs** and take **BP, pulse and temperature**. BP **160/110 or more** (or 140/90 with symptoms), shock index **0.9 or more**, fever **38.0°C or more**, bleeding, fits or labour → she skips the queue and the emergency pathway starts.
25	A single 1+ glycosuria at ANC: GDM?	**Not by itself.** The renal threshold for glucose falls in pregnancy, so isolated glycosuria is often physiological. Repeat the dipstick, check RBS, and do a **75 g OGTT at 24–28 weeks** if it persists or she has risk factors. Diagnose GDM only on blood glucose.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 22-year-old primigravida attends her first ANC contact at a HC III at 16 weeks by LNMP. She is HIV-negative, well, and has no drug allergies. Which set of interventions should she receive TODAY?

A Iron–folic acid only; start IPTp at 20 weeks
B IPTp-SP only, as the other drugs can wait until 20 weeks
C Iron–folic acid, first dose of IPTp-SP (DOT), Td1, mebendazole 500 mg, LLIN, and booking tests
D Iron–folic acid, Td1 and cotrimoxazole prophylaxis
E Booking tests only; treatment starts once results are back

2. A 27-year-old G3P2 at 24 weeks, known HIV-positive on TLD and cotrimoxazole 960 mg daily, attends ANC. The midwife asks whether to give IPTp-SP. What is the correct advice?

A Do not give SP; cotrimoxazole already provides malaria protection and combining sulfa drugs increases toxicity
B Give SP today and monthly until delivery
C Give SP but stop cotrimoxazole for 1 week
D Give SP only if a malaria RDT is positive
E Replace cotrimoxazole with SP for the rest of the pregnancy

3. A 19-year-old G1 at 28 weeks has Hb 8.4 g/dL. She feels tired but has no breathlessness at rest. BP 110/70 mmHg, pulse 92 bpm, temperature 37.9°C. What is the MOST appropriate next step?

A Transfuse 2 units of packed cells today
B Reassure her; this is physiological haemodilution
C Start folic acid 5 mg daily alone
D Malaria test, blood film and stool microscopy, and start therapeutic iron (about 120 mg elemental iron daily) with folic acid
E Give intravenous iron immediately

4. A 30-year-old G2P1 at 34 weeks has BP 148/96 mmHg, repeated after rest as 146/94 mmHg. At her 26-week visit BP was 118/74 mmHg. She has no headache or visual symptoms. What is the most important immediate investigation at the ANC clinic?

A Obstetric ultrasound
B Urine dipstick for protein
C Random blood sugar
D Malaria RDT
E Serum uric acid

5. At a 36-week ANC contact a 25-year-old G2P1 has a firm, round, ballotable mass at the fundus and a soft broad mass above the pelvic brim. FHR is 144 bpm above the umbilicus. What is the BEST plan?

A Reassure; the baby will turn spontaneously in labour
B Book her for a home delivery with a traditional birth attendant
C Plan vaginal breech delivery at the HC III
D Induce labour today
E Confirm by ultrasound and refer to hospital for ECV or planned caesarean section

6. A 26-year-old G1 at 30 weeks has a fundal height (SFH) of 25 cm. Dating was by a 10-week ultrasound. She has chronic hypertension. What is the MOST likely explanation and next step?

A Wrong dates; repeat LNMP history
B Fetal growth restriction; arrange ultrasound for biometry, liquor and umbilical artery Doppler
C Normal variation; review in 4 weeks
D Polyhydramnios; check blood sugar
E Twin pregnancy; refer for scan

7. A 23-year-old G1 at 20 weeks has a reactive rapid syphilis test and a positive confirmatory test. She has no symptoms and does not know when she was infected. She has no allergies. What is the correct treatment?

A Benzathine penicillin G 2.4 MU IM single dose
B Amoxicillin 500 mg 8-hourly for 7 days
C Ceftriaxone 250 mg IM single dose
D Doxycycline 100 mg 12-hourly for 28 days
E Benzathine penicillin G 2.4 MU IM weekly for 3 weeks, and treat her partner

8. A 21-year-old woman books at 10 weeks. Which of the following is NOT appropriate at this visit?

A Iron–folic acid
B Td vaccine
C HIV, syphilis and HBsAg testing
D Ultrasound to confirm dates and viability
E First dose of IPTp-SP

9. A 34-year-old G4P3 at 12 weeks had severe pre-eclampsia requiring delivery at 32 weeks in her last pregnancy. BP today is 120/78 mmHg. Which preventive intervention is MOST important to start now?

A Low-dose aspirin 75 mg (75–150 mg) at night from now until about 36 weeks
B Methyldopa 250 mg 8-hourly
C Magnesium sulphate prophylaxis
D Bed rest at home until 28 weeks
E Salt restriction

10. A 28-year-old G2P1 at 32 weeks attends the ANC clinic saying the baby has not moved since yesterday. What should you do FIRST?

A Book an ultrasound for next week
B Ask her to count kicks for 12 hours at home
C Auscultate the fetal heart (Doppler or Pinard) now and assess the mother's BP and urine
D Give IPTp and reassure
E Induce labour immediately

11. A 24-year-old HIV-negative woman at 26 weeks received her first dose of IPTp-SP 3 weeks ago at 23 weeks. She comes today for review. What is the correct action regarding SP?

A Give SP today because it is a scheduled visit
B Give a double dose to catch up
C Stop IPTp because she already had one dose
D Postpone SP until at least 4 weeks after the last dose (next week or next contact)
E Switch to weekly chloroquine prophylaxis

12. A 17-year-old primigravida, height 146 cm, books at 22 weeks at a HC II. All her booking tests are normal. Where should she be advised to deliver?

A At home with a trained TBA, because she is low-risk
B At the HC II with the midwife
C At a HC III, which is the nearest facility
D It does not matter as long as she attends ANC
E At a HC IV or hospital that can perform caesarean section

13. A 32-year-old G3P2 at 35 weeks has Hb 6.2 g/dL, is breathless on mild exertion, and has pulse 112 bpm and bilateral basal crepitations. What is the BEST immediate management?

A Admit, sit her up, give oxygen, and transfuse packed red cells slowly with furosemide 20–40 mg IV; look for and treat the cause
B Oral ferrous sulphate 200 mg 8-hourly and review in 2 weeks
C Intravenous iron infusion as an outpatient
D Emergency caesarean section to cure the anaemia
E Rapid transfusion of 3 units of whole blood

14. A 29-year-old G2P1 at 12 weeks is O Rh-negative. Her partner is Rh-positive. She had a normal delivery 2 years ago at home and received no injections afterwards. What is the most useful next step?

A Give anti-D now and every month
B Nothing; Rh disease only affects first pregnancies
C Check for anti-D antibodies (indirect Coombs test) and plan anti-D after any sensitising event and after delivery
D Terminate the pregnancy
E Give IV immunoglobulin weekly

15. A 26-year-old G1 at 41 weeks + 1 day by a first-trimester scan attends ANC. She is well, with a cephalic presentation, FHR 136 bpm and normal BP. What is the most appropriate plan?

A Wait until 44 weeks for spontaneous labour
B Offer induction of labour in a facility that can do caesarean section
C Immediate caesarean section
D Send her home and review in 2 weeks
E Castor oil at home

Answers & explanations

> **1. C** · Iron–folic acid, first dose of IPTp-SP (DOT), Td1, mebendazole 500 mg, LLIN, and booking tests
>
> At 16 weeks she is past the first trimester, so she qualifies for **IPTp-SP (from 13 weeks)**, **deworming** (after the first trimester), **Td1**, daily **iron–folate** and an **LLIN**, alongside booking tests. ANC aims to deliver the whole package at every contact to avoid missed opportunities. A delays SP unnecessarily. B omits iron, Td and deworming. D: cotrimoxazole is for HIV-positive women and would contraindicate SP. E wastes the contact; the preventive drugs do not depend on results.

> **2. A** · Do not give SP; cotrimoxazole already provides malaria protection and combining sulfa drugs increases toxicity
>
> Uganda MoH and WHO guidance: **HIV-positive women on cotrimoxazole should not receive SP**. Cotrimoxazole (a sulfonamide–antifolate like SP) gives antimalarial protection, and the combination risks **severe cutaneous reactions and marrow toxicity**. B and C ignore this. D: a positive RDT needs **treatment** (artemether–lumefantrine), not IPTp. E: cotrimoxazole is continued for its opportunistic-infection benefits.

3. D · Malaria test, blood film and stool microscopy, and start therapeutic iron (about 120 mg elemental iron daily) with folic acid

Hb 8.4 g/dL is **moderate anaemia** (7–9.9). Physiological haemodilution rarely takes Hb below 10.5–11 (B is wrong). She has a low-grade fever, so **malaria must be excluded**. Look for the **cause** (film, malaria, hookworm) and start **therapeutic iron** (e.g. ferrous sulphate 200 mg twice daily ≈ 120 mg elemental iron) with folic acid. A: transfusion is for severe anaemia (below 7 g/dL near term or decompensated). C: folic acid alone does not treat iron deficiency. E: IV iron is for intolerance, non-response or late presentation.

4. B · Urine dipstick for protein

New hypertension after 20 weeks is gestational hypertension or **pre-eclampsia**. The key immediate bedside test that separates them is **urine protein**. Proteinuria (or organ dysfunction) makes it pre-eclampsia, which needs admission. A (growth scan) and E (uric acid) are helpful later but do not change today's triage. C and D are not relevant to the new hypertension.

5. E · Confirm by ultrasound and refer to hospital for ECV or planned caesarean section

The findings suggest **breech presentation** at 36 weeks. Spontaneous version becomes unlikely after 36 weeks (A). Breech needs a **hospital birth**: offer **ECV** (if no contraindication) or a **planned CS**. Ultrasound also checks placental site and anomalies. B and C are unsafe. D: there is no indication for induction.

6. B · Fetal growth restriction; arrange ultrasound for biometry, liquor and umbilical artery Doppler

SFH is **5 cm below** dates, and the dates are reliable (early scan). With **chronic hypertension**, placental insufficiency makes **FGR** the leading diagnosis. Arrange a **growth scan with liquor and Doppler**. A: dates were confirmed early. C: a difference of more than 3 cm is not normal variation. D and E cause a **large-for-dates** uterus.

7. E · Benzathine penicillin G 2.4 MU IM weekly for 3 weeks, and treat her partner

Latent syphilis of **unknown duration** is treated as late latent: **benzathine penicillin G 2.4 MU IM weekly × 3**, and treat the partner. A is for early syphilis (under 2 years). B and C do not reliably treat syphilis. D: **doxycycline is contraindicated in pregnancy** (teeth and bone) and does not treat the fetus.

8. E · First dose of IPTp-SP

SP is **not given in the first trimester**. Pyrimethamine is a folate antagonist, and IPTp starts from **13 weeks**. All the other options are standard at an early booking. Early booking is ideal because it allows an accurate dating scan and early folate.

9. A · Low-dose aspirin 75 mg (75–150 mg) at night from now until about 36 weeks

Previous pre-eclampsia is a **high-risk factor**. **Low-dose aspirin** started **before 20 weeks** (ideally from 12) reduces recurrence by improving placentation and lowering thromboxane. Calcium is also useful if dietary intake is low. B: she is normotensive, so antihypertensives are not indicated. C: $MgSO_4$ is for severe pre-eclampsia or eclampsia. D and E do not prevent pre-eclampsia.

10. C · Auscultate the fetal heart (Doppler or Pinard) now and assess the mother's BP and urine

Reduced fetal movements are a **danger sign** that may precede stillbirth. **Check the fetal heart immediately** and assess the mother for causes (pre-eclampsia, fever, abruption). If the heart is absent, confirm IUFD by ultrasound; if abnormal, refer for monitoring. A and B delay care. D misses the problem. E is premature without assessment.

11. D · Postpone SP until at least 4 weeks after the last dose (next week or next contact)

IPTp-SP doses must be **at least 4 weeks (1 month) apart**. Only 3 weeks have passed, so schedule the next dose once 4 weeks have elapsed. A breaks the interval rule. B is unsafe. C: the target is **3 or more doses**. E: chloroquine prophylaxis is not recommended because of resistance.

12. E · At a HC IV or hospital that can perform caesarean section

Two risk factors, **age below 18** and **height below 150 cm**, raise the risk of **cephalopelvic disproportion and obstructed labour**. She needs delivery where **CS and blood** are available (HC IV or hospital). A, B and C lack CS capacity. D is false: ANC attendance does not replace an appropriate birth plan.

13. A · Admit, sit her up, give oxygen, and transfuse packed red cells slowly with furosemide 20–40 mg IV; look for and treat the cause

This is **severe anaemia with cardiac decompensation**. She needs admission and slow transfusion of **packed cells** (less volume), with a **diuretic** to prevent pulmonary oedema, and investigation and treatment of the cause (malaria, worms, HIV, sickle cell). B and C are too slow for decompensation. D: delivery does not treat anaemia and she would tolerate PPH badly. E: rapid whole-blood transfusion risks fatal fluid overload.

14. C · Check for anti-D antibodies (indirect Coombs test) and plan anti-D after any sensitising event and after delivery

She may already be **sensitised** (no anti-D after her last delivery). Check an **antibody screen (indirect Coombs)**. If negative, give **anti-D within 72 hours** of sensitising events and after delivery of an Rh-positive baby (plus routine antenatal prophylaxis where available). If positive, refer for monitoring of fetal anaemia. B is the reverse of the truth: **later** pregnancies are at risk. A, D and E are inappropriate.

15. B · Offer induction of labour in a facility that can do caesarean section

WHO recommends **induction of labour from 41 weeks** in a low-risk pregnancy with reliable dating, because stillbirth and perinatal morbidity rise beyond this. It should be done at a facility with CS capacity. A and D expose her to post-term risks. C: there is no indication for CS. E is unsafe and unproven.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The late booker: catching up on everything

Nakato, a 20-year-old G1P0 from Mukono, books at a HC IV at 30 weeks. She says she did not come earlier because she had no transport money and feared the HIV test. She has no complaints.

1. What history do you want, and why?

> **LNMP** and certainty (dating); **danger signs** (bleeding, headache, fever, leaking); fetal movements; **medical history** (sickle cell, hypertension, heart disease); drug and **sulfa allergy**; any previous tests or ANC elsewhere; **social**: support, partner, IPV screen, transport, money for delivery. Her age (under 20) is itself a risk factor.

2. Examination findings: BP 124/78 mmHg, mild pallor, SFH 29 cm, singleton, longitudinal lie, cephalic, 5/5 palpable, FHR 140 bpm. Urine: protein negative. What tests do you send today?

> **Hb**, **blood group and Rh**, **HIV and syphilis** (dual rapid test), **HBsAg**, urinalysis (done), and **malaria RDT** given the pallor. Request an **ultrasound** (dating is now less accurate at 30 weeks, but it confirms a singleton, placental site, liquor and growth).

3. Results: Hb 9.6 g/dL, HIV negative, syphilis negative, HBsAg negative, RDT negative. What do you give her today?

> The **catch-up package**: **IPTp-SP 3 tablets DOT** (first dose); **Td1** 0.5 mL IM (Td2 in 4 weeks); **mebendazole 500 mg** single dose; **therapeutic iron** (ferrous sulphate 200 mg twice daily ≈ 120 mg elemental iron + folic acid 400 µg) because she has moderate anaemia; **LLIN**. Counsel on diet (beans, greens, meat, vitamin C with iron; no tea with tablets).

4. How do you counsel her, and what is her birth plan?

> Danger signs (use teach-back). Birth plan: she is **under 20 and a primigravida**, so she should deliver at the **HC IV** (CS capable). Identify a companion, transport number, savings, blood donor and mama kit. Discuss postpartum family planning and exclusive breastfeeding. Explain why the HIV test matters and that it will be **repeated in the third trimester or labour**.

5. Plan her remaining contacts.

> Contacts at **34, 36, 38 and 40 weeks**: BP and urine every time; **IPTp doses 2 and 3** at 34 and 38 weeks (at least 4 weeks apart); Td2 at 34 weeks; **Hb recheck** at 34 weeks (expect a rise of about 1 g/dL per 2–3 weeks); HIV retest; presentation at 36 weeks; **induction if undelivered at 41 weeks**.

Take-away: A late booker has missed many opportunities. Deliver the whole package at the first contact (tests, IPTp, Td, deworming, iron, net), treat what you find, and schedule the remaining third-trimester contacts carefully.

Case 2: Routine visit, silent danger

Apio, a 35-year-old G5P4 at 33 weeks, comes for a routine ANC contact at a HC III. She says she is fine but has 'a little headache' since morning. Her previous BP readings were normal.

1. What will you check immediately, and why?

> **BP** (repeat after rest), **urine protein**, and symptoms of severe disease: headache severity, **visual disturbance**, **epigastric/RUQ pain**, vomiting, reduced fetal movements. Also check reflexes (clonus), oedema of the face and hands, and FHR. A headache at 33 weeks is **pre-eclampsia until proven otherwise**.

2. BP 164/112 mmHg on repeat, proteinuria 3+, with brisk reflexes. What is the diagnosis and what are your immediate steps at the HC III?

> **Pre-eclampsia with severe features.** Call for help. Left lateral position; IV access; send FBC if possible; urinary catheter. **$MgSO_4$ loading dose: 4 g IV slowly over 5–10 minutes (20 mL of 20%) + 10 g IM (5 g of 50% in each buttock)**. Antihypertensive: **nifedipine 10 mg orally** (repeat per protocol) or **hydralazine 5 mg IV** slowly. **Refer urgently** to the HC IV or hospital with a health worker, and phone ahead.

3. What should ANC have picked up earlier, and what risk factors does she have?

> Risk factors: **age 35 or over**, grand multiparity (P4 going on 5). She did not have a high-risk factor for aspirin by herself, but BP and urine should be checked at every contact, and new symptoms should be sought. Grand multiparity also raises her risk of **PPH and malpresentation**, so she should deliver in hospital.

4. At the hospital, what definitive management follows?

> Continue **$MgSO_4$ 5 g IM 4-hourly** (alternate buttocks) for 24 hours after delivery or the last fit, with monitoring (RR at least 16/min, patellar reflexes present, urine at least 30 mL/h; antidote **calcium gluconate 1 g IV**). Control BP. Give **dexamethasone 6 mg IM 12-hourly × 4** for fetal lung maturity (below 34 weeks), and plan **delivery once stable** (severe pre-eclampsia at 33 weeks: deliver after stabilisation). Use active management of the third stage (oxytocin 10 IU IM), **avoiding ergometrine**.

5. What follow-up does she need after delivery?

> BP monitoring (antihypertensives weaned over days to weeks), review at 6 weeks, family planning (she is grand multiparous; offer long-acting methods or tubal ligation if she desires), and counselling that she has a higher risk of pre-eclampsia next time (**aspirin from 12 weeks**) and of future cardiovascular disease.

> **Take-away:** BP and urine at every contact are non-negotiable. A 'little headache' in the third trimester is pre-eclampsia until proven otherwise; stabilise with $MgSO_4$ and antihypertensives before referral.

Case 3: New HIV diagnosis at booking

Nabirye, a 26-year-old G2P1 at 14 weeks, books at Mengo Hospital ANC. Her dual HIV/syphilis rapid test is HIV-reactive and syphilis-non-reactive. Her first child, now 3 years old, has never been tested.

1. How do you confirm and communicate the result?

> Follow the **national HIV testing algorithm** (a confirmatory test with a second assay) before diagnosis. Give **post-test counselling** in private: explain the result, reassure her that with ART **transmission to the baby can be reduced to below 1–2%**, and address disclosure fears and IPV risk.

2. What treatment do you start and when?

> **Same-day ART**: **TLD** (tenofovir 300 mg / lamivudine 300 mg / dolutegravir 50 mg) **one tablet daily**. Start **cotrimoxazole 960 mg daily** per national criteria, which means **no IPTp-SP**. Continue iron–folate, Td1, deworming (she is past the first trimester) and an LLIN. Screen for **TB** symptoms.

3. What tests and monitoring follow?

> Baseline: HBsAg (TLD covers HBV too), creatinine if available, CD4 (advanced disease screen: CrAg if CD4 is low). **Viral load** per the eMTCT schedule (about 3 months after starting ART, and again in late pregnancy/at delivery). Adherence counselling at each visit.

4. Who else needs testing?

> **Index testing**: her **partner** (couple testing and PrEP if he is negative and she is not yet suppressed) and her **3-year-old child**. Children of positive mothers must be tested regardless of age.

5. What is the plan for delivery and the baby?

> Facility delivery. Avoid unnecessary ARM or episiotomy. The baby gets **infant ARV prophylaxis** (nevirapine ± zidovudine according to risk, per national guidelines), **DNA PCR at 6 weeks** (EID), and cotrimoxazole from 6 weeks until HIV is excluded. **Exclusive breastfeeding** for 6 months while the mother continues ART. Postpartum family planning (TLD has no important interactions with implants).

Take-away: ANC is the gateway to eMTCT: confirm, counsel and start TLD the same day. Remember that cotrimoxazole replaces SP, and test the partner and existing children.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.